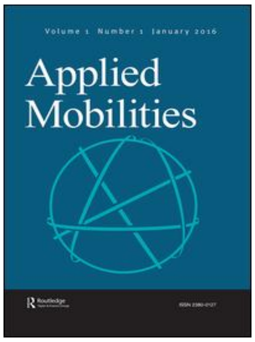

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ARTICLE

Mobility justice and accessible public transport networks for people with intellectual disability

Ellen van Holstein [a], Ilan Wiesel [a] and Crystal Legacy [b]

[a]School of Geography, University of Melbourne, Melbourne, Australia; [b]Melbourne School of Design, Melbourne, Australia

ABSTRACT

This paper aims to advance the debate on transport accessibility. While cognitive barriers to accessibility receive some attention in public transport research and policy, studies focus on discrete services and little research has addressed access solutions focused on the scale of the transport network. To address this gap, this paper presents the results from a one-day focus group with 16 public transport and disability advocacy practitioners in Victoria, Australia. The study reveals three concurrent ongoing developments in Victorian public transport that pose both opportunities and barriers to improving accessibility for people with intellectual disabilities across the extended metropolitan network. They are: diversification of transport services and providers; increased reliance on innovation in communication technology; and the need for inclusive forms of citizen participation mechanisms. The paper builds on the theory of mobility justice to argue that achieving greater transport accessibility requires network-wide, collaborative policies on accessibility. The paper demonstrates that a focus on cognitive access barriers in public transport enriches mobility justice as it highlights the distributed responsibility across a public transport network for the limited mobility of a marginalised group.

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1. Introduction

One of the normative goals guiding public transport planning is to support and expand transport accessibility across cities and regions (Curtis and Scheurer 2016; Wittowsky, Van der Vlugt, and Nadler 2019). Yet, barriers remain in governance, regulatory, and planning systems to achieving accessibility ambitions, and this is especially true for accessibility planning for people with intellectual disability (ID). This is a critical gap. With population ageing in many western countries, the number of people with a cognitive impairment is growing (Anstey 2016). Slowly, researchers give attention to this group, highlighting the need for consistency in communication and adequately trained staff (Risser, Iwarsson, and Ståhl 2012; Risser et al. 2015; Bigby et al. 2019). However, this research is currently limited to highlighting the potential of technological innovation (Stock et al. 2011), the value of well-trained and disability aware staff (Tillmann et al. 2013), and accessibility of separate services for people with particular impairments (Bigby et al. 2019). Public transport

CONTACT Ellen van Holstein ellen.van@unimelb.edu.au

systems offer greater mobility to passengers who have the skill and cognitive capacity to combine multiple modes of transport on one trip (Oostendorp et al. 2019), and this potentially excludes passengers with ID. This paper answers a call for attention to intermodality in accessibility research, put forward by Oostendorp et al. (2019) in this journal. It does so by analysing how collaboration, innovation and co-production in public transport systems as a network consisting of different modes, routes and providers, can support communication accessibility. Using Melbourne's extended public transport system and its privatised and franchised public transport network as a case study, we illustrate challenges to overcoming access barriers, with particular attention given to station and urban design based barriers such as unclear wayfinding and inconsistent messaging.

In 2012, the Australian Bureau of Statistics (ABS) estimated that approximately 3% of Victoria's population has an ID, defined broadly as "reduced ability to understand new or complex information, and to learn and apply new skills" (Australian Bureau of Statistics (ABS) 2012). This definition includes people with developmental disabilities, as well as cognitive disabilities that occur later in life such as acquired brain injury or dementia. Public transport services interact with injury and impairment in disabling ways. To illustrate, close to 60% of people with ID have severe communication limitations (AIHW 2008) and literacy is a common difficulty for people with ID. Meanwhile, written information continues to be the most common form of communication about service delivery in public transport, causing notable access barriers. While some transport service providers in Victoria have started to pay attention to communication barriers in their policy and practice (Bigby et al. 2019), these initiatives have focused on separate services with the aim of upgrading each to a "communication accessible" status. This paper acts on the momentum generated by this increasing awareness to highlight the importance of network-wide collaboration on communication accessibility. It demonstrates the accessibility benefits that transport service providers can achieve when they share resources such as communication standards, staff training and their experiences organising inclusive transport user committees.

This paper builds on the conceptual framework of "mobility justice" put forward by Sheller (2018) to develop holistic, network wide approaches to promoting the social inclusion of people with ID through opportunities to participate in public transport. It does so by reporting on a focus group discussion about creating an accessible public transport network for people with ID between public transport professionals and disability advocates in Melbourne in the State of Victoria, Australia. Focus group participants included representatives from key transport service providers and disability advocacy groups. Analysis of the discussion highlighted the need to consider communication accessibility in the context of developments in public transport networks and the wider context that shapes people's mobility. Specifically, analysis revealed three concurrent ongoing developments in Melbourne's public transport network that pose opportunities and barriers to improving its accessibility for people with ID: 1) the increased reliance on technologies to communicate transport information; 2) service integration across diverse transport services and providers; and 3), the development of citizen participation mechanisms such as consumer feedback groups in transport planning and policy development.

In the following section we discuss conceptualisations of disability and mobility. We offer details on focus group methods to proceed to the results. We conclude with an analysis of the importance of collaborative approaches for achieving communication accessible public transport systems.

2. The mobility justice framework

Public transport and accessibility remain central priorities in transport research and policy, as illustrated by key publications (Curtis and Scheurer 2016; Martens 2016, 2018) as well as the recent special issue on accessibility in this journal (Wittowsky, Van der Vlugt, and Nadler 2019). The transport justice framework goes someway to link space and mobility in discussions about accessibility. However, it tends to overlook how people are differently embodied and how the interactions between the physical environment including transport infrastructure, and people who are differently gendered, racialized and enabled are part of politics of movement that reach beyond the transport system. The ongoing interest in accessibility coincides with the spatial turn and the new mobilities paradigm in social sciences. This turn has infused the new transport agenda with critical perspectives on the meaning of key concepts such as accessibility and justice (Sheller and Urry 2006, 2016; Sheller 2018).

The mobility justice framework put forward by Sheller (2018) considers how politics of movement are shaped by the discursive construction of white, masculine and enabled bodies as idealised performers of mobile subjectivities (also see Sheller and Urry 2006; Cresswell 2010). This view reveals how the mobility of some are celebrated while others' are policed or received with suspicion. The discursive context in which opportunities for mobility and their appropriations take shape are particularly important to understand the barriers people with ID experience. As people with ID are frequently excluded from employment opportunities, they are almost by default overlooked by public transport policies that focus on commutes. Thus, while people with disability generally receive insufficient priority in public transport provision (e.g. Sawchuk 2014; Goggin 2016; Sheller 2018), people with ID are even further marginalised within that group. In addition to a long history of exclusion from employment and mainstream community spaces, this invisibility is arguably deepened by reliance on disability support staff for overcoming barriers. While independence and spontaneity feature centrally in modern constructions of meaningful movement (e.g. Curtis and Scheurer 2016; Endres, Manderscheid, and Mincke 2016), for people with ID reliance on pre-booked support staff limits mobility and spontaneity in their social lives (Milner and Kelly 2009).

A mobility justice framework thus pays attention to the social construction of appropriate forms of mobility for differently positioned people, and concurrently, what constitutes accessibility for people who are differently enabled. Because the framework recognises that mobility injustices emerge beyond the confines of transport infrastructure and are shaped by unequal power relationships in wider society, it recognises the importance of the inclusion of marginalised groups in decision-making processes that shape opportunities for mobility (Sheller 2018). The expertise of non-traditional transport users such as poor people and people with disability, allows the introduction of new problems and ideas to conversations about transport, mobility and accessibility. For these reasons, this paper introduces accessibility challenges for people with ID into the mobility justice framework and argues for greater public transport accessibility and inclusion in decision-making for people with ID.

3. Cognitive barriers in public transport

In Australia, services for people with disability are now funded through a new, individualised funding model called the National Disability Insurance Scheme (NDIS). People with

ID are the largest group of NDIS participants. A fundamental aim of the NDIS is to enhance and support people with disabilities' participation in mainstream services (Wiesel et al. 2019). For example, NDIS participants may be eligible for budgets for specialist transport assistance only if they "cannot use public transport without substantial difficulty due to their disability" (NDIS 2019). The complexity of public transport systems and their rushed environments make people with ID more reliant on support and people spend a high proportion of individual support packages on transport costs, mostly in the form of taxis and support staff (National Disability Services 2018). The inaccessibility of public transport for people with ID is thus an increasingly pressing social and economic issue.

Mobility justice research has analysed the mobility experiences of people with physical and sensory impairments, but the experiences of people with ID have mostly featured in transport research (Risser, Iwarsson, and Ståhl 2012) and in disability focused disciplines such as social work (Wasfi, Steinmetz-Wood, and Levinson 2017; Trescher 2018). While narrower in focus than work with a mobility lens, this small but important body of work offers insight into the barriers experienced by people with ID and has highlighted potential solutions.

Research literature on the travel experiences of people with conditions such as ID, stroke, dementia and dyslexia indicate a pattern of reoccurring challenges for this diverse group of passengers (Rosenkvist et al. 2010; Risser, Iwarsson, and Ståhl 2012; Lamont, Kenyon, and Lyons 2013; Bigby et al. 2019). Service users with communication support needs, including people with ID, report unpredictable journeys (Bigby et al. 2019). Delays, platform changes, and encounters with staff with different communication styles are examples of components that lead to unpredictability. Unpredictability undermines travel routines and makes independent travel harder for people with ID. Moreover, passengers with dyslexia and cognitive challenges reported difficulties when confronted with large amounts of information, and when having to retain information to act on later (e.g. Rosenkvist et al. 2010; Lamont, Kenyon, and Lyons 2013). How transport providers communicate information thus creates cognitive tasks that form barriers to service use for people with ID.

Research reveals solutions within and beyond transport systems. Solutions located at the scale of the individual passenger include interventions such as travel training and assistive technologies such as transit apps (e.g. Livingstone-Lee, Skelton, and Livingston 2014). Another set of attempts to address the inaccessibility of public transport systems is the provision of "community" transport options, which segregate people with support needs into parallel transport systems. As suggested by Miciukiewicz and Vigar (2012: 1943) this "drift towards a 'single problem, single answer' has held back a richer understanding of the multiple ways in which transport and mobility issues are implicated in broader processes of social cohesion". While these interventions can have important effects on the mobility and independence of individuals, focus on the individual reduces incentives to make transport systems structurally accessible (Martens 2018).

Solutions within the reach of public transport providers include staff awareness training (Tillmann et al. 2013) structural adjustments to the ways they communicate to patrons, and the roll-out of system-wide technologies such as hearing loops or trip-planning apps. Researchers stress the importance of messages that are limited to the most important words, and that are offered when and where passengers must make decisions (Risser, Iwarsson, and Ståhl 2012). Various forms of legislation and theoretical frameworks acknowledge the importance of legible, clear and consistent information for the accessibility of public transport systems (e.g. Australian Government 2002). Yet, detailed information on the travel experiences of passengers with ID, and on opportunities for improvement, remain scarce (Rosenkvist et al. 2010; Risser, Iwarsson, and Ståhl 2012; Lamont, Kenyon, and Lyons 2013; Bigby et al. 2019). When they do exist, they are often limited to particular transport modes, rather than considering a whole transport system (Risser et al. 2015).

Victoria is considered a world leader in accessible public transport (Bigby et al. 2019). While most investments have targetted physical access, communication access increasingly also receives attention. Although transport providers have long recognised the importance of plain language and clear communication, communication accessibility is a relatively new concept that researchers introduced to remove barriers to information and communication that impede effective and uncompromised use of services (Johnson et al. 2013). Communication access is being promoted by disability support provider "Scope" who use staff training, customer assessments and awards to make transport services more inclusive (Solarsh and Johnson 2017; Bigby et al. 2019). Scope works with service providers on communication features such as drop-in opportunities for face-to-face interaction, the presence of communication resources at service counters and staff trained to use them, and accessible written information and signage. Communication accessibility recognises that people communicate in various ways: through symbols, routine, gestures, facial expressions, verbal communication and written word. Of these, written and verbal forms of communication are often experienced as complex ways of communication by people with cognitive disabilities. Yet, services typically rely on these two ways to provide information (Hartley Kean 2016). As a result, service use is obstructed or challenging for a considerable portion of the population.

Discrete service providers are engaging with communication access and this has increased accessibility for people with ID. However, transport over long distances requires the use of multiple modes of transport offered by different providers (Risser, Iwarsson, and Ståhl 2012; Oostendorp et al. 2019). Without collaboration across the public transport sector, initiatives that promote accessibility remain fragmented. Collaboration between industry partners, enables consistent experiences of access for service users with diverse needs throughout their journey (Risser, Iwarsson, and Ståhl 2012). For this reason, this paper applies the concept "communication access" to understand ways the accessibility of Melbourne's public transport system can be improved for people with ID.

4. Method

This paper presents the results of a focus group on accessible public transport for people with ID. Focus group discussion is a qualitative research method that can bring together a group of people with similar experiences, backgrounds or expertise. While focus groups do not allow generalisations, they lend themselves to unearthing experiences and opinions in a group (Cameron 2016). Participants of the focus group in our study were selected based on their expertise on the accessibility of public transport services in Melbourne. The group represented the perspectives and experiences of service providers, transport users and transport staff. Van Holstein developed a list of the largest public transport providers and other relevant stakeholders such as the Public Transport Users'

Association. She then sought representatives with expertise on accessibility and inclusion working at these organisations through internet searches and authors' contacts. This led to additional identification of a not-for-profit that provides volunteer-based assistance to travellers and the union that represents public transport staff.

Recruitment and snowballing led to 16 focus group participants. Participants included two inclusion and accessibility officers as well as one general manager who together represented the three largest bus and rail services operating on the Melbourne network; one inclusion officer for a regional rail service that connects the metropolitan area to regional cities; the inclusion officer and a way-finding expert from Public Transport Victoria (PTV), which is the Victorian Government statutory authority that manages Victoria's public transport network and its electronic ticketing system; a representative from the Transport Bus Rail Union; the president of the Public Transport Users' Association; and the manager of a not-for-profit that provides trained volunteer services to people who need assistance while traveling. Two participants were representatives from an advocacy organisation for people with ID, and one of these representatives contributed his lived expertise of ID to the discussion.

The focus group served to scope and promote an industry wide approach to inclusive public transport for people with ID. The focus group was designed as a performative research initiative (Cameron, Manhood, and Pomfrett 2011), meaning that it aimed to record current initiatives, while it simultaneously promoted stakeholder awareness and encouraged collaboration. Before roundtable discussions started, experts from Scope gave a presentation about communication accessibility. This way, the roundtable discussion occurred in the current momentum for thinking about access barriers for people with ID. The needs and potential for greater independence of people with ID remain overlooked and the three structured discussions encouraged participants to explore possibilities for applying communication accessibility initiatives to meet the needs of people with ID. We structured the focus group discussion around the following questions which Wiesel and Legacy presented to the group over three collective sessions:

(i) What current initiatives are under way to advance accessibility of transport services for people with ID?
(ii) What barriers and opportunities do you see for the promotion of accessibility for this group in the transport system?
(iii) Are inclusive mechanisms in place for service user participation in decision-making, and if not, how might these be created?

Participants answered these questions individually at first and the focus group leader subsequently moderated open conversation that responded to the questions. Some participants asked other participants for further detail or explanation to clarify responses and positions. When disagreement emerged, for example between the union representative and transport service providers, the focus group leaders ensured participants had equal opportunity to voice their thoughts and concerns and both views were recorded to ensure inclusivity of all views. A notetaker recorded the discussions and the authors analysed the notes thematically.

5. Public transport in Melbourne

Melbourne's public transport network includes train, tram and bus services that show limited physical integration. Some bus services feed into a radial rail network, but for a large part, people need to rely on walking, cycling, car and bus services to connect to tram and train services (Infrastructure Victoria 2016). The lack of physical integration is mirrored in the fragmented ownership and governance of public transport services, a phenomenon shaped by privatisation described by Graham and Marvin (2001) as "splintered urbanism". Public transport in Victoria is a highly deregulated market. The turn to neoliberal policies and governance arrangements in the 1990s saw the privatisation of Melbourne's expansive public transport network in 1999. While the state retained ownership of rail infrastructure, the operation of trams, suburban trains and buses was privatised. Private operators compete for leases through a franchising model, which places responsibility for the operation and maintenance of assets on franchisees (Ashmore, Stone, and Kirk 2019). Operators compete for patrons rather than strategically plan and coordinate across the different modes of transport, and the result is a poor service for all, including for those who rely strongest on public transport (Mees, 2009).

Melbourne's splintered transport system has caused fragmented accessibility. As public transport users rely on multiple service providers for longer journeys, passengers must adapt to different communication styles, fonts and delivery of visual and audio messages on these trips. The distributed responsibility in the transport system requires collaboration between transport service providers. It also requires a view on transport service experience that considers relationships between services in passengers' travel chains (Risser, Iwarsson, and Ståhl 2012).

At the time when the focus group was conducted, at an institutional level, the Minister of Public Transport carried the state's responsibility for public transport. As the minister's statutory body, PTV aimed to integrate different services into a coherent and user-friendly public transport system. PTV integrated services by combining information from different providers into one trip planner, the PTV app. PTV has also introduced an electronic ticketing system that integrates the payment systems of separate providers. While the new payment system has created uniformity, the complexity of an electronic travelcard also raised issues for people with ID. Following a campaign by disability rights activists, PTV exempted some people with disability from travel fares to sidestep this complexity. PTV also works to integrate the accessibility of services in the network. For example, PTV coordinates a Public Transport Access Committee (PTAC) that is appointed by the Minister of Public Transport to assess the accessibility of the public transport network. This committee consists of people with professional or lived experience of sensory and/or physical disability who the Minister selects based on expressions of interest and interviews. This committee complements the customer reference groups of transport providers that give feedback on separate services.

PTV ensures that franchisees comply with state and federal legislation, including anti-discrimination legislation and accessibility guidelines. Federal policy dictates explicit disability access standards with which providers must comply. *The Disability Standards for Accessible Public Transport*, (the standards), is the guiding document for accessible public transport in Australia (Australian Government 2002). It was issued in 2002 in response to the Disability Discrimination Act of 1992. The standards state transport providers' obligations under the act. Since 2002, amended versions of the standards have dictated very specific accessibility requirements such as the width of ramps, the visibility of signage and the font size of printed information.

While accessible formats and easy to understand English have been part of the standards since 2002, cognitive access barriers have started to emerge on agendas only recently. The Department of Infrastructure and Regional Development recently published "The Whole Journey" in response to the second review of the standards (The Department of Infrastructure and Regional Development 2017). The guide recognises that providers' mere compliance with the standards will not make whole journeys accessible, for example because the environment outside stations or integrated services information are inaccessible. The Department reports that "people with disability have highlighted that they are less likely to embark on public transport journeys that involve interchanges as it adds complexity and uncertainty to their trip" (Department of Environment, Land, Water and Planning 2017, 36). Institutional fragmentation remains an important and critical barrier to overcoming inaccessibility that requires coordinated and collaborative governance and leadership across different agencies and departments (Legacy, Curtis, and Sturup 2012). Because of the challenges identified in the research literature around intermodality, decision-making under time pressure and unexpected changes in platforms, the focus group was designed to scope opportunities for an industry wide approach to accessibility. This way the focus group was developed to contribute to making journeys more consistent, predictable and therefor easier to navigate.

6. Trajectories towards accessibility: integration, innovation and participation

The results sections follow three concurrent developments that shape accessibility of public transport systems: technological innovation, integration of fragmented systems, and fostering participation in decision-making. These three developments featured prominently in focus group participants' answers to questions about adjustments that are underway to improve communication accessibility and to further the accessibility of the public transport network for people with ID.

6.1. From fragmentation to integration

Fragmentation of the public transport system, that manifests in challenges such as poorly aligned stopovers, poses severe barriers for people with ID. Many accessibility solutions that participants discussed fitted in with efforts to create an integrated public transport system. Uniformity in wayfinding design was one suggested way to establish consistency, for instance, a focus group participant affiliated with PTV discussed current developments in colour-coded wayfinding design. Consistency in colour coding throughout the public transport system can create easy to learn and easy to follow routes through stations and public transport nodes.

Participants discussed style guidelines as an opportunity to develop consistent communication and design standards across services. PTV's journey planner allows passengers to plan their trip across different modes, including walking to and from stops and stations. According to focus group participants, PTV style guidelines have improved the consistency of signage across modes. Further opportunities for integration lie with the training and instruction of ground staff. Technological innovation might make the availability and communication of information to public transport users more effective, but many will continue to prefer to engage with staff either by default or in times of disruption.

Participants discussed a program called "Traveling in the Shoes of Others" run by PTV. In this program, staff are tasked with navigating the system with an ascribed handicap. For instance, a tram conductor is instructed to go to a destination without using their voice or handheld devices. Focus group participants discussed opportunities to adapt the program to heighten staff awareness about passengers with ID. They envisioned that a collaborative program shared by different service providers will generate more consistent awareness among staff working across the public transport system.

Participants offered that another way to overcome the effects of fragmentation is a "no wrong door"-approach, whereby all staff can assist customers using any service offered at one station. This discussion revisited the importance of good staff working conditions and workloads for improving accessibility. The working conditions of transport staff do not receive enough attention in transport and mobilities research, even though the experiences of workers are co-constitutive of the accessibility and affordability of transport networks (Rekhviashvili and Sgibnev 2018). A union representative lamented that vital services that enable intermodal travel for passengers with needs are offered by volunteers rather than paid staff. The participant highlighted that volunteers provide a valuable service, but that transport providers' responsibilities for accessibility must not be placed on philanthropic organisations and volunteers, particularly where this previously provided paid work. Australia has an ageing population, and the amount of people who rely on help to navigate a complex and increasingly crowded environment is expected to grow. More structural and robust solutions must be found to address the increasing skills that are required to navigate public transport services.

6.2. Diversity and integration in innovative technological solutions

Focus group discussions about current and necessary innovations to make the Victorian network more accessible consistently touched on technological innovation, which, we argue, must also be considered as an aspect of network-wide accessibility. Providers increasingly communicate routine information about public transport services via websites and mobile apps. For instance, the PTV trip-planner app has become particularly important as an integration measure as it combines timetables of all services. In reflecting on the useability of those apps, participants engaged in a discussion about the value of and barriers caused by mobile apps. Disability advocates offered that apps are frequently updated, and sometimes in ways that require users to change how they locate information. According to them, the frequent and extensive updating of mobile apps makes them unsuitable for people who need to invest time to learn to navigate new technology. They also discussed that not all people have phones, know how to, or are able to, access or zoom in on timetables, which creates barriers to service use (e.g. Bigby et al. 2019). They stressed that it is important that information remains available and up-to-date on a website as well. This part of the focus group demonstrated that it is important for accessibility that integration is not achieved monolithically by the introduction of one app, but that it engages diverse technologies and formats.

Transport service providers communicate information about services differently at times of disruption, and real-time information is more readily available at stations than on websites and apps. Unexpected changes can trigger anxiety in people with limited experience or confidence in independent travel. Therefore, focus group participants discussed communication of unexpected changes. Focus group participants saw a potential solution in a low-tech itinerary or travelcard for passengers with disability to be used across the transport system. The passenger or a support person can fill out this travelcard before travel with information such as the destination, planned and alternative transfers, and abilities and barriers to keep in mind when helping to reroute this passenger. Passengers can present the card to ground staff of different transport services in times of disruption who will then know how best to assist.

Several participants affiliated with a service provider reported that their organisations organise "Try Before You Ride" events to reach out to disengaged service users. They reported that these events are opportunities for potential public transport users to become familiar with stations and vehicles in a low-pressure, static environment. They allow people to become familiar with the location and functions of buttons which differ on different types of vehicles in a fleet. The opportunity to familiarise oneself with infrastructure can help reinstate confidence in people who might have experienced stressful travel. This initiative resonates with the insight that experiences of fear and lack of confidence have a strong influence on people's use of public transport and thus their mobility (e.g. Risser, Iwarsson, and Ståhl 2012; Lamont, Kenyon, and Lyons 2013). An intermodal, holistic approach to improve accessibility highlights the need for collaboration across multiple providers in designing initiatives such as "Try Before You Ride" where passengers are also able to experience and practice intermodal transfers across services. Focus group participants discussed options to use virtual reality technology to make people familiar with vehicles, stops, stations and platforms. It is a good way for people to experience a full journey involving multiple transport modes at a slow pace and in a controlled way. Service providers in the focus group discussed the challenge to reach people who do not use public transport, such as people with ID, because they are unlikely to be part of groups that are targeted for these events. However, the reported success of "Try Before You Ride" events indicates that this might be one way to support people with ID to use public transport.

Participants discussed beacon technology as a potential innovation that makes stations more accessible. Participants reported a recent trial with this technology to give people with low-vision directional advice at stations. They suggested this technology might be further developed to allow passengers to access information about a vehicle such as estimated arrival time, current passenger load relative to capacity, priority seating, and assistance and door buttons. These developments will allow passengers to better prepare for boarding and this might reduce nervousness about journeys. Focus group participants suggested that government support for this technology is pivotal, and that an open source system would allow roll-out throughout the system.

The focus group sparked debate about the relationships between staff and technological innovation in the creation of accessible services. Participants considered that some people with disability dislike disclosing their disability to get the support or information they need (e.g. Bigby et al. 2019). Some people with disability prefer technological solutions and some prefer face-to-face communication which requires the continuing presence of drop-in opportunities. Focus group participants agreed that staff and technological solutions should both be considered when maximising accessibility. A union representative highlighted the importance of well-supported staff for achieving accessibility. During times of disruption, many service users require information and there is less opportunity for staff to accommodate people who need more time or assistance to adjust their travel plans. The focus group discussed whether understaffing and resulting pressure on staff in the context of service disruptions reduces the availability of staff and compromises their ability to communicate service information patiently and effectively. Inclusion officers of service providers rebuffed any accusations of increasing workloads or deteriorating working conditions with the argument that new technologies are developed to support rather than replace service providers' ground staff.

According to service providers it would be beneficial if different public transport providers could integrate knowledge about where their staff are and allow the communication of this information to public transport users. Although members of staff are often present on trains, trams and stations, they might not always be visible or approachable for passengers, for example, because they are traveling on the next carriage. Apps or websites can direct passengers to the nearest members of staff and heighten the effectiveness of existing staff's presence. According to these focus group participants, this will promote the ability of passengers to ask questions about where to alight or transfer, and to confirm that one is traveling in the right direction. For people who are not comfortable speaking up, or who have to work up to asking questions, the technology might help passengers seek the attention of staff, for example by offering a chat messenger service. The discussion thus embraced the realisation that technology alone cannot overcome the disabling barriers of transport systems (Gleeson 2005).

6.3. Participation

People with ID are excluded from public transport and participation in community life, and for the same reasons they are less likely to be included in political and institutional decision making about public transport (e.g. Bascom and Christensen 2017). Participation in public transport in the fullest sense means being able to use services without limitations, but also to be included in shaping public transport – at the scale of both discrete services and networks – on an equal basis with other users. Participatory mechanisms such as deliberative committees are commonplace in public services and are vital for ensuring community support for transport planning and service delivery (Legacy 2018). Existing models for the participation of people with or without disability in government and corporate decision-making processes tend to favour the most skilled and vocal. Researchers have also identified these issues in mechanisms designed to include people with ID in deliberative planning (Frawley and Bigby 2011). During the focus group, public transport services representatives indicated that they use various committees and reference groups to inform decisions regarding service delivery. Some of these committees have an explicit focus on disability, access or inclusion. The focus group revealed that service providers experience similar challenges which might be addressed most effectively with an integrated, collaborative approach.

Collaboration across diverse providers has the potential to enhance participation of people with ID in decision-making across the network. The discussion revealed some integration in approaches to participation of people with disability as various representatives indicated that they turn to PTAC for advice on disability inclusion and access. In addition to this central body for advice and feedback on disability access, service providers also run their own consumer consultation committees. Like PTAC, these committees tend to focus on sensory and physical disabilities, and often do not include people with ID. As a result of the discussion, participants recognised the need to pro-actively include people with ID in consultation processes. They suggested that they might do this by adding people with ID to existing accessibility committees or by setting up separate committees to represent existing and prospective users with ID.

Participants discussed various challenges they might face in their attempts to set up such committees as well as possible solutions. Participants indicated that it is difficult to find people who are able to represent particular groups, such as people with ID. Participants reported that some public transport providers have a database containing contact information of people with various access needs and that this database is valuable for ad hoc consultation on changes that are being made in a short time frame. Participants were unsure whether the database included people with ID. To assist, a disability advocate offered to help find suitable candidates for reference groups. Disability advocacy groups can help recruit and support committee members, and review disability actions plans and service changes that are likely to affect people with ID. Disability advocacy groups provide a wealth of insight into the travel and service experience of public transport users with ID. Additionally, this moment of exchange and the initiative it inspired illustrates the performative value of our focus group research method for transport accessibility and mobility justice

People with ID are a hard-to-reach group and meaningful participation requires support and capacity building (Frawley and Bigby 2011). Collaboration across providers is an effective way to build capacity for both participants with disabilities, and for services facilitating reference groups. A group of experienced people with ID participating in several reference groups run by different providers can be useful for building the necessary skills for meaningful participation among a wider community of people with ID. Providers sharing information about their engagement of users with ID in reference groups is also critical to building meaningful participation. Currently, the isolation of people with disabilities and the limited skills in this group means that the same experienced individuals with ID are invited to reference groups (Frawley and Bigby 2011). Collaboration between services and between the public transport and disability advocacy sector might allow coordination of capacity building and spread the consultation workload and opportunity across people with ID.

The focus group discussion explored the appropriateness of consultation and highlighted that this should be considered when developing access committees and reference groups. Disability advocates offered that a committee appointment takes considerable time and service providers must value and acknowledge a committee member's contribution. A participant with disability contributed that too often, people with disability are asked to contribute without pay. It is important that people are renumerated for the time and expertise they contribute to meetings. He highlighted that too often, people with disability are not paid for their expertise, and this unequal treatment can be eliminated by renumeration through salaries, or alternatively gift vouchers if financial compensation is not possible. Disability advocates expanded that for some, vouchers may be preferable since they do not impact on welfare payments. For others, payment through vouchers does not fully convey recognition of vocational advocacy work. A coordinated approach can help establish appropriate standards of renumeration across a public transport system.

7. Discussion and conclusion

While scholarly debates in mobility studies highlight the need to understand the power relationships that shape inequal mobilities (Cresswell 2010; Sheller 2018), focus on disability in this field has been limited to the barriers people with physical and sensory impairments experience. As a consequence, researchers have largely overlooked the governance and leadership required to overcome barriers caused by cognitive tasks. Cognitive barriers are caused by demands to read, engage with technologies, process large amounts of information, etc., that are embedded in increasingly splintered transport landscapes (however see Oostendorp et al. 2019). The focus group presented in this paper has used the concept of communication accessibility to consider cognitive access barriers holistically as part of a mobility justice approach. This approach has revealed that a focus on ID, as an overlooked group of people who experiences limited mobility due to circumstances inside and outside public transport systems, enriches the mobility justice framework.

A focus on ID contributes a heightened awareness of access barriers caused by a splintered transport system in which private lease holders communicate differently (Graham and Marvin 2001). These access barriers reveal the value of approaches based on collaboration inside and beyond the public transport network to include disability support providers and staff representative bodies. The research demonstrates that the focus group is a performative research method that "is geared towards crafting rather than capturing realities" (Cameron, Manhood, and Pomfrett 2011, 493). In other words, not only did the focus group record accessibility measures in the transport system, it also encouraged practitioners to think about accessibility in multi-scalar and intermodal ways that go beyond the constituency of the discrete service they provide. This performative approach has the potential to make the mobility justice actionable in research practice.

The research brought service providers, state authority and disability advocates together in a focus group that used a mobility justice framework to interpret mobility outcomes as generated at the intersection of the transport system and wider social circumstances (Sheller 2018). A focus on ID revealed that collaborations with the disability support sector can lead to structural changes in the public transport network and demonstrated that these changes can reach maximum impact when applied by all providers. Structural changes create opportunities for accessibility and inclusion that do not solely rely on individualised disability support staff. Initiatives to accommodate the needs of people with disability outside the main service, for example when services offer volunteer assistance or alternative modes of transport, will stall structural improvements in accessibility to that service (Martens 2018). Focus on cognitive access barriers in mobility justice thus highlights the distributed responsibility of providers to address barriers structurally.

Discussion about ID and cognitive access barriers created some awareness about contexts that shape the ability of people with disabilities to use different transport options, such as funding and availability of support staff and taxis. This highlights that a mobility justice approach requires intermodal collaboration. Discussion indicated that technological innovation is most likely to lead to consistency and predictability in customers' experiences when it is paralleled with an ongoing commitment to staff training and the protection and promotion of favourable working conditions and workloads. Rapid changes in the transport industry such as the introduction of rideshare companies are creating pressure on existing service providers, and in turn on staff, in the form of increasing workloads and precarious working conditions (Rekhviashvili and Sgibnev 2018). Focusing on cognitive tasks and communication accessibility highlights the importance of abundant, high-skilled staff and reveals alliances in transport systems between staff and service users, for instance around protecting workloads and training opportunities, that can be leveraged for greater mobility justice.

The focus group discussion also revealed that there is an ongoing need to create awareness amongst providers of the need to engage directly with people with ID. The conversation revealed a broadly carried consensus on the value of customer engagement but also revealed that people with ID remain overlooked when composing representative bodies. Participation is an important aspect of mobility justice (Sheller 2018), and a focus on ID reveals that a holistic approach also needs to consider that the ability to participate is unequally distributed. The approach in this research reveals that transport providers need to create some capacity among their staff and their customers with disability to achieve broader mobility justice goals. A network-broad commitment to include people with ID service planning will make public transport more accessible for people with ID and for a growing group of people who acquire cognitive disabilities later in life.

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ORCID

Ellen van Holstein http://orcid.org/0000-0002-7244-1775
Ilan Wiesel http://orcid.org/0000-0003-2049-5883

Crystal Legacy http://orcid.org/0000-0002-8687-7297

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